HENGKAI YE

Ph.D. Candidate
College of Information Sciences and Technology
The Pennsylvania State University

hengkai@psu.edu
765-772-8417
https://hengkai-ye.github.io/
E338 Westgate Building
University Park, PA 16802

RESEARCH INTERESTS

Software and System Security: Focus on uncovering novel attack surfaces and exploitation techniques in software and system, with the goal of designing robust and comprehensive defenses.

AI Security: Investigate, evaluate, and mitigate security risks in LLM-based applications. Leverage LLMs to enhance traditional security tasks such as malware detection.

EDUCATION

The Pennsylvania State University, State College, PA, USA — May 2022 - Dec 2026 (expected)

Ph.D. in Informatics

Dissertation: *Understanding and Identifying Security-Critical Data for Attacks and Defenses*

Advisor: Hong Hu

Purdue University, West Lafayette, IN, USA — Aug 2020 - May 2022

M.S. in Computer and Information Technology

Huazhong University of Science and Technology, Wuhan, China — Sept 2016 - Jun 2020

B.E. in Information Security

EMPLOYMENT

Sr. Staff Security Researcher, Palo Alto Networks, Santa Clara, CA, USA — May 2026 - Present

Enhancing WildFire, an advanced malware prevention engine, to reduce false-positive and false-negative rates.

Research Assistant, The Pennsylvania State University, State College, PA, USA — May 2022 - May 2026

PUBLICATIONS

Refereed Conference Proceedings

[1] **ContractGuard: Auditing Semantic Contracts of MCP Tools for Security Violations**.
Hengkai Ye*, Zhechang Zhang*, Ruibo Lu, Jinyuan Jia, and Hong Hu.
In *Proceedings of the ACM Conference on Computer and Communications Security (**CCS**)*, The Hague, The Netherlands, November 2026.

[2] **NCFuzz: Configuration-guided Network Service Fuzzing**.
Xuesong Bai, Hengkai Ye, Shenghan Zheng, Fenglu Zhang, Hong Hu, and Zhou Li.
In *Proceedings of the ACM International Symposium on Software Testing and Analysis (**ISSTA**)*, Oakland, CA, October 2026.

[3] **SACK: Systematic Generation of Function Substitution Attacks Against Control-Flow Integrity**.
Zhechang Zhang, Hengkai Ye, and Hong Hu.
In *Proceedings of the Annual Network and Distributed System Security Symposium (**NDSS**)*, San Diego, CA, February 2026.

[4] **Too Subtle to Notice: Investigating Executable Stack Issues in Linux Systems**.
Hengkai Ye and Hong Hu.
In *Proceedings of the Annual Network and Distributed System Security Symposium (**NDSS**)*, San Diego, CA, February 2025.

[5] **VIPER: Spotting Syscall-Guard Variables for Data-Only Attacks**.
Hengkai Ye, Song Liu, Zhechang Zhang, and Hong Hu.
In *Proceedings of the USENIX Security Symposium (**Security**)*, Anaheim, CA, August 2023.

[6] **BET: black-box efficient testing for convolutional neural networks**.
Jialai Wang, Han Qiu, Yi Rong, Hengkai Ye, Qi Li, Zongpeng Li, and Chao Zhang.
In *Proceedings of the ACM International Symposium on Software Testing and Analysis (**ISSTA**)*, July 2022.

Journal Articles

[7] **Interpreting Deep Learning-based Vulnerability Detector Predictions Based on Heuristic Searching**.
Deqing Zou, Yawei Zhu, Shouhuai Xu, Zhen Li, Hai Jin, and Hengkai Ye.
In *ACM Transactions on Software Engineering and Methodology (**TOSEM**)*, 2021.

Others

[8] **Executable Stack Issues in Software Transformation (Accepted As Talk)**.
Hengkai Ye and Hong Hu.
In *Workshop on Forming an Ecosystem Around Software Transformation (**FEAST**)*, October 2024.

[9] **One Flip is All It Takes: Identifying Syscall-Guard Variables for Data-Only Attacks**.
Hengkai Ye, Hong Hu, Song Liu, and Zhechang Zhang.
In *Black Hat Asia Briefings*, April 2024.

Preprint

[10] **TrustDesc: Preventing Tool Poisoning in LLM Applications via Trusted Description Generation**.
Hengkai Ye, Zhechang Zhang, Jinyuan Jia, and Hong Hu.
In *arXiv Preprint*, 2026.

Under Review

[11] **FLAG: Semantics-Driven Identification of Security-Critical Data**.
Zhechang Zhang*, Hengkai Ye*, Ruibo Lu, and Hong Hu.

HONORS AND AWARDS

Student Travel Grant, USENIX Security	2023
Mingde Scholarship, Huazhong University of Science and Technology	2019
Freshman Scholarship, Huazhong University of Science and Technology	2016

ACTIVITIES

Poster Program Committee Member

ACM Conference on Computer and Communications Security (CCS)	2025

Artifact Evaluation Committee Member

Network and Distributed System Security Symposium (NDSS)	2025, 2026

External Reviewer

USENIX Security Symposium (Security)	2025, 2026
Network and Distributed System Security Symposium (NDSS)	2023, 2024, 2025, 2026
ACM Conference on Computer and Communications Security (CCS)	2022, 2024, 2025
Applied Cryptography and Network Security (ACNS)	2026

Research in Attacks, Intrusions and Defenses (RAID) 2026

Reviewer

[Springer Journal] Cybersecurity

Teaching Assistant

IST 454 Computer and Cyber Forencis Fall 2022

OPEN SOURCE CONTRIBUTION

Sack: A pipeline for constructing end-to-end SUB attacks under precise CFI protection [3]
https://github.com/PSU-Security-Universe/sack

BadAss: Demonstration of the BadAss issue [4]
https://github.com/PSU-Security-Universe/badass

Viper. A tool for identifying syscall-guard variables [5]
https://github.com/PSU-Security-Universe/viper

Data-only attacks. A list of data-only attacks [5]
https://github.com/PSU-Security-Universe/data-only-attacks